Measuring the gap between the Melnikov threshold and the period-doubling cascade in the damped driven pendulum

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2026-07-10 · Pendulum Lab numerical laboratory (TypeScript engine, externally cross-validated)

Abstract. The damped driven pendulum $\ddot{\theta} = -\sin\theta - \gamma\dot{\theta} + A\cos(\omega t)$ ($\omega = 0.6667$) carries two distinct, frequently conflated notions of a "chaos threshold": the analytic Melnikov amplitude $A_c(\gamma)$, above which first-order perturbation theory predicts a transverse homoclinic tangle, and the period-doubling onset $A_{PD}(\gamma)$, where the primary period-1 attractor begins the Feigenbaum cascade that produces the sustained chaotic attractor. We measure A_{PD} over $\gamma \in [0.1, 0.8]$ with an attractor-strobed bisection refined by the Floquet multiplier of the Newton periodic orbit (onset interpolated at $\rho = -1$), and compare it with the closed-form A_c. The ratio A_{PD}/A_c falls monotonically from 2.38 at $\gamma = 0.1$ to 0.987 at $\gamma = 0.8$, and the widely quoted ordering $A_c < A_{PD}$ *reverses* near $\gamma \approx 0.69$: at low damping the tangle precedes the cascade by a wide and rapidly growing margin, while at strong damping the cascade begins *below* the first-order Melnikov prediction. At the literature point $\gamma = 0.5$ our measurement $A_{PD} = 1.0664$ agrees with the published 1.0663 (Baker & Gollub) to four digits. A reduced-grid frequency scan ($\omega = 0.5, 0.85$) and a Duffing double-well companion map show the same monotone gap closure — with the crossing damping itself moving with drive frequency — so the reversal is a robust property of the first-order threshold, not an artifact of the classic parameter point.

1. Introduction

The sinusoidally driven, damped pendulum is the canonical low-dimensional route to chaos, and it supports two different analytic/numerical landmarks as the drive amplitude A grows. The Melnikov method gives a closed-form first-order threshold $A_c = (4\gamma\omega_0/\pi)\cdot\cosh(\pi\omega/2\omega_0)$ above which the stable and unstable manifolds of the hilltop saddle intersect transversally, creating a Smale-horseshoe tangle. The tangle guarantees *transient* chaos and fractal basin boundaries — not a chaotic attractor. The sustained chaotic attractor instead appears at the end of a period-doubling cascade of the primary period-1 response, whose onset A_{PD} is a property of a specific attractor branch and has no closed form. Textbooks typically note $A_c < A_{PD}$ at the classic parameter point γ = 0.5, ω = 2/3. This paper asks the quantitative question: *how does the gap between the two thresholds behave as damping is varied?*

2. Methods

All computations use the open Pendulum Lab engine, whose equations of motion are validated component-wise against an independent SymPy symbolic derivation (max relative deviation ≈ 10^{-14}) and whose trajectories are cross-validated against SciPy DOP853 at rtol = 10^{-13}; the engine also reproduces published anchors (elliptic pendulum period, double-pendulum normal modes, and the γ = 0.5 period-doubling onset). The drive is made autonomous by carrying the phase as a third coordinate; integration is RK4 with dt = 0.005 snapped so an integer number of steps spans one drive period exactly.

A_{PD} measurement. For each γ, the drive amplitude is marched upward from $0.9 \cdot A_c$ with the strobed state warm-started at every step, so the measurement follows the physically realised attractor branch (this matters: the symmetric period-1 orbit pitchforks before the cascade, and a Newton continuation of the *symmetric* orbit would miss the doubling of the symmetry-broken branch the attractor actually follows). The loss of period-1 stability is bisected on the strobe map (transients of 300–600 drive periods; period detected in the (sin θ, cos θ, ω) embedding, immune to 2π winding). The bracket is then refined by Newton periodic orbits *seeded from the attractor*: the most negative real Floquet multiplier ρ(A) of the orbit is interpolated through ρ = −1. A measurement is accepted as a period doubling only when this crossing is verified; otherwise the loss is reported as non-PD. Halving dt changes the γ = 0.5 onset by |Δ| ≈ 7.1e-13, so discretisation error is negligible at the quoted precision.

Corroboration. At $0.97 \cdot A_{PD}$ and $1.08 \cdot A_{PD}$ the Gottwald–Melbourne 0–1 test is applied to the cos θ strobe series (700 samples): K ≈ 0 confirms regular motion below onset; the value above onset depends on whether $1.08 \cdot A_{PD}$ lands beyond the cascade accumulation point or inside a periodic window, and is reported without prejudice.

3. Results

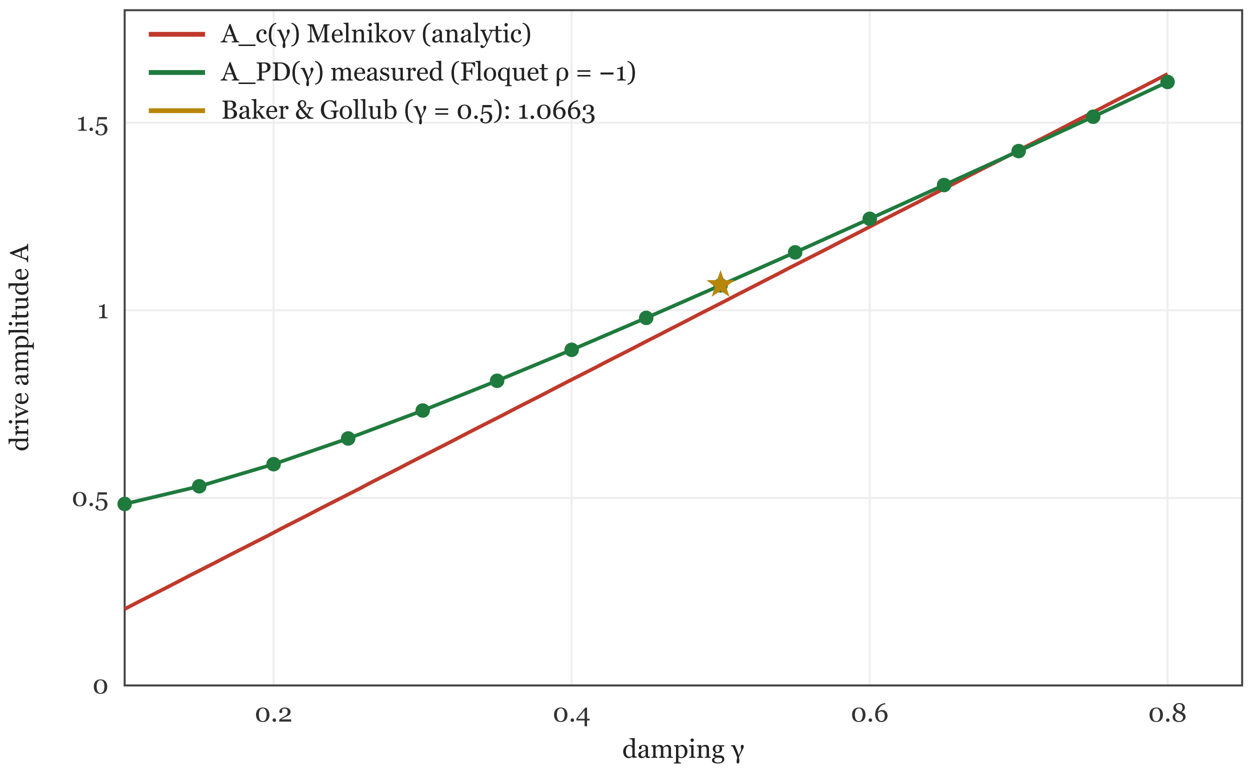

Figure 1. The analytic Melnikov threshold $A_c(\gamma)$ (red line) and the measured period-doubling onset $A_{PD}(\gamma)$ (green circles, Floquet-refined). The star is the published γ = 0.5 value 1.0663 (Baker & Gollub); our measurement at that point is 1.06637. A_c is exactly linear in γ; the measured cascade onset is not. Certification artifact: `reports/flagship-figure1.svg`, SHA-256 prefix `07f877d6fdb816`; numerical rows and uncertainty brackets are cross-referenced in Appendix A.

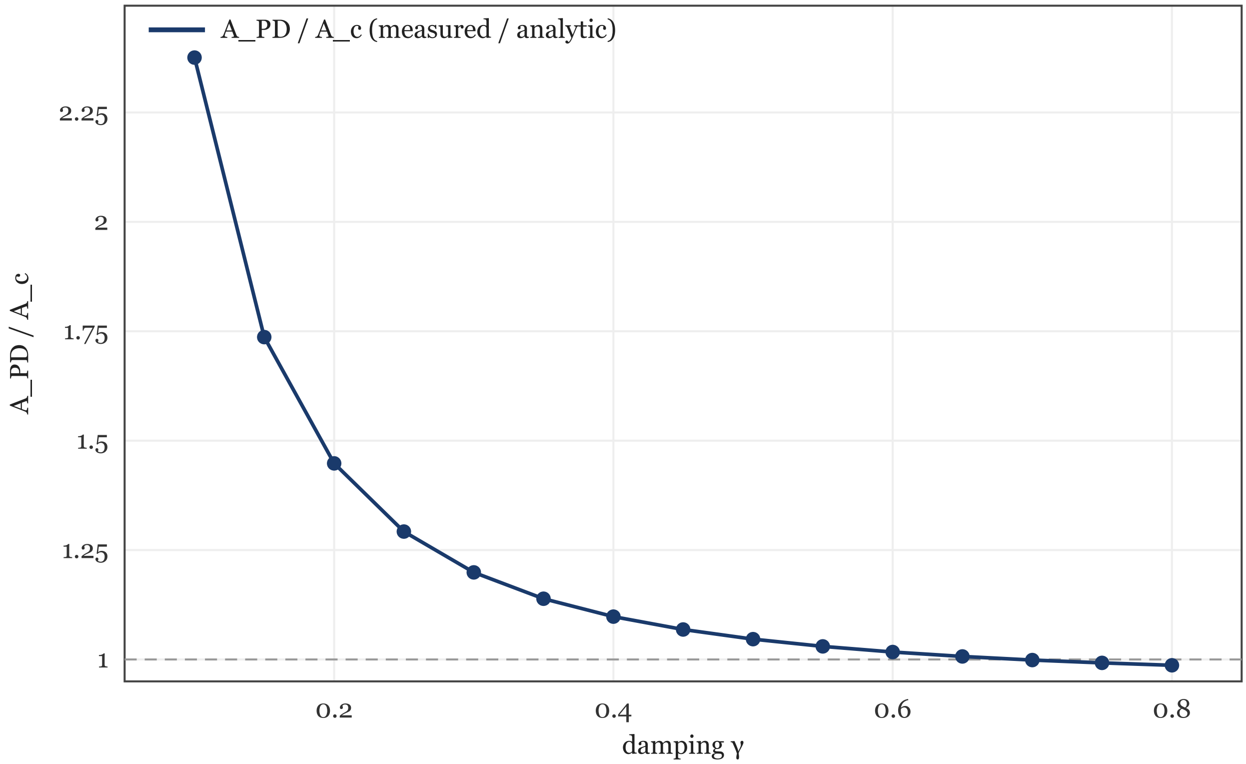

Figure 2. The ratio A_{PD}/A_c versus damping. The gap closes monotonically with increasing γ and crosses 1 near γ ≈ 0.69: beyond this damping the period-doubling cascade of the primary attractor begins *below* the Melnikov tangle threshold. At γ = 0.1 the cascade requires 138% more drive than the tangle.

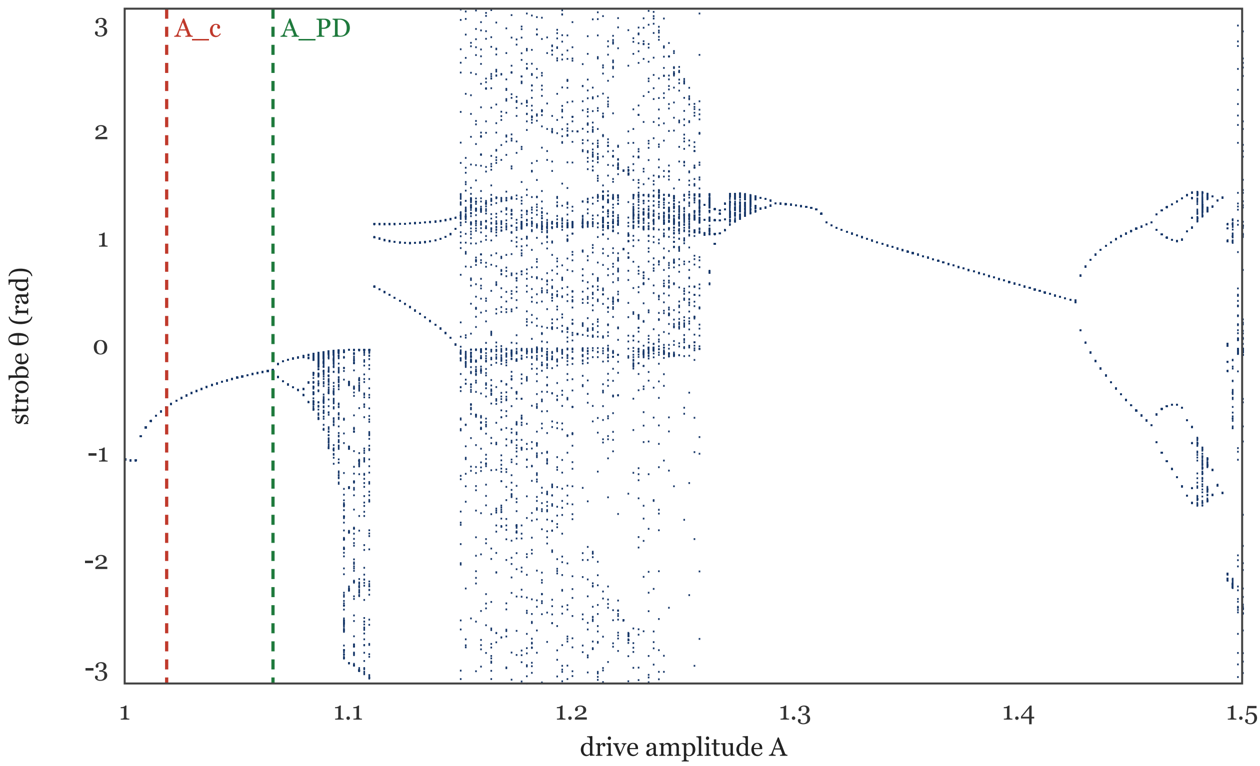

Figure 3. Strobe bifurcation diagram at $\gamma = 0.5$ (θ sampled once per drive period after a 250-period transient, warm-started in A). The dashed lines mark A_c (red) and the measured A_{PD} (green). Between them the motion is periodic but the phase space already contains the homoclinic tangle: transient chaos and fractal basin boundaries without a strange attractor. The cascade, chaotic band, and the large periodic (rotating) window are visible beyond A_{PD}.

γ	A_c (Melnikov)	A_PD (measured)	A_PD/A_c	loss of period-1	K at 0.97·A	K at 1.08·A
0.10	0.20375	0.48398	2.3753	PD (ρ → −1)	0.02	1.00
0.15	0.30563	0.53080	1.7367	PD (ρ → −1)	0.02	1.00
0.20	0.40751	0.59004	1.4479	PD (ρ → −1)	0.02	-0.01
0.25	0.50939	0.65829	1.2923	PD (ρ → −1)	-0.02	1.00
0.30	0.61126	0.73294	1.1991	PD (ρ → −1)	0.02	0.06
0.35	0.71314	0.81217	1.1389	PD (ρ → −1)	-0.02	1.00
0.40	0.81502	0.89470	1.0978	PD (ρ → −1)	0.02	-0.01
0.45	0.91690	0.97964	1.0684	PD (ρ → −1)	-0.02	-0.01
0.50	1.01877	1.06637	1.0467	PD (ρ → −1)	-0.02	1.00

γ	A_c (Melnikov)	A_PD (measured)	A_PD/A_c	loss of period-1	K at 0.97·A	K at 1.08·A
0.55	1.12065	1.15448	1.0302	PD (ρ → −1)	-0.02	1.00
0.60	1.22253	1.24368	1.0173	PD (ρ → −1)	-0.01	1.00
0.65	1.32441	1.33377	1.0071	PD (ρ → −1)	0.02	-0.01
0.70	1.42628	1.42463	0.9988	PD (ρ → −1)	0.02	-0.57
0.75	1.52816	1.51622	0.9922	PD (ρ → −1)	0.02	0.13
0.80	1.63004	1.60853	0.9868	PD (ρ → −1)	0.02	1.00

4. Discussion

Three regimes emerge. (i) **Low damping ($\gamma \lesssim 0.2$):** $A_c \to 0$ linearly while the cascade onset of the primary resonance remains an order-one amplitude, so the ratio diverges (2.38 already at $\gamma = 0.1$). The window of "tangle but no strange attractor" — long chaotic transients and fractal basin boundaries below a still-periodic attractor — is widest here, and the phase space is visibly multistable (see §5). (ii) **Moderate damping:** the textbook ordering $A_c < A_{PD}$ holds, but the margin shrinks steadily (to 5% at $\gamma = 0.5$). (iii) **Strong damping ($\gamma \gtrsim 0.69$):** the measured cascade begins *below* the first-order Melnikov prediction. This is not a contradiction — the Melnikov threshold is asymptotically exact only as $\gamma, A \to 0$, and by $\gamma \approx 0.7$ the perturbation parameter is $O(1)$ — but it sharpens the usual caveat into a measured boundary: the first-order formula stops being even an ordering bound near $\gamma \approx 0.69$.

The 0–1 test values corroborate the structural picture: $K \approx 0$ on the period-1 side everywhere, while above onset K depends on where $1.08 \cdot A_{PD}$ falls relative to the cascade accumulation point and the periodic windows visible in Figure 3 — both outcomes occur in the table, as expected for a Feigenbaum scenario with embedded windows.

5. Beyond one frequency and one system

Two extensions probe whether the closing gap is an artifact of the classic parameter point. First, a reduced-grid **frequency scan** repeats the full Floquet-refined measurement at $\omega = 0.5$ and $\omega = 0.85$. The shape survives: the ratio falls monotonically with damping at every frequency, and the crossing damping γ^* moves with ω — at $\omega = 0.85$ the gap is still open at $\gamma = 0.8$ (ratio 1.096), while at $\omega = 0.5$ the measured points at $\gamma = 0.65$ and 0.8 already sit *below* 1 (0.941, 0.927). At ω = 0.5, γ = 0.2; ω = 0.5, γ = 0.35; ω = 0.5, γ = 0.5 the primary branch loses period-1 stability without a verified ρ = −1 crossing, and those points are reported as unclassified rather than forced into the map.

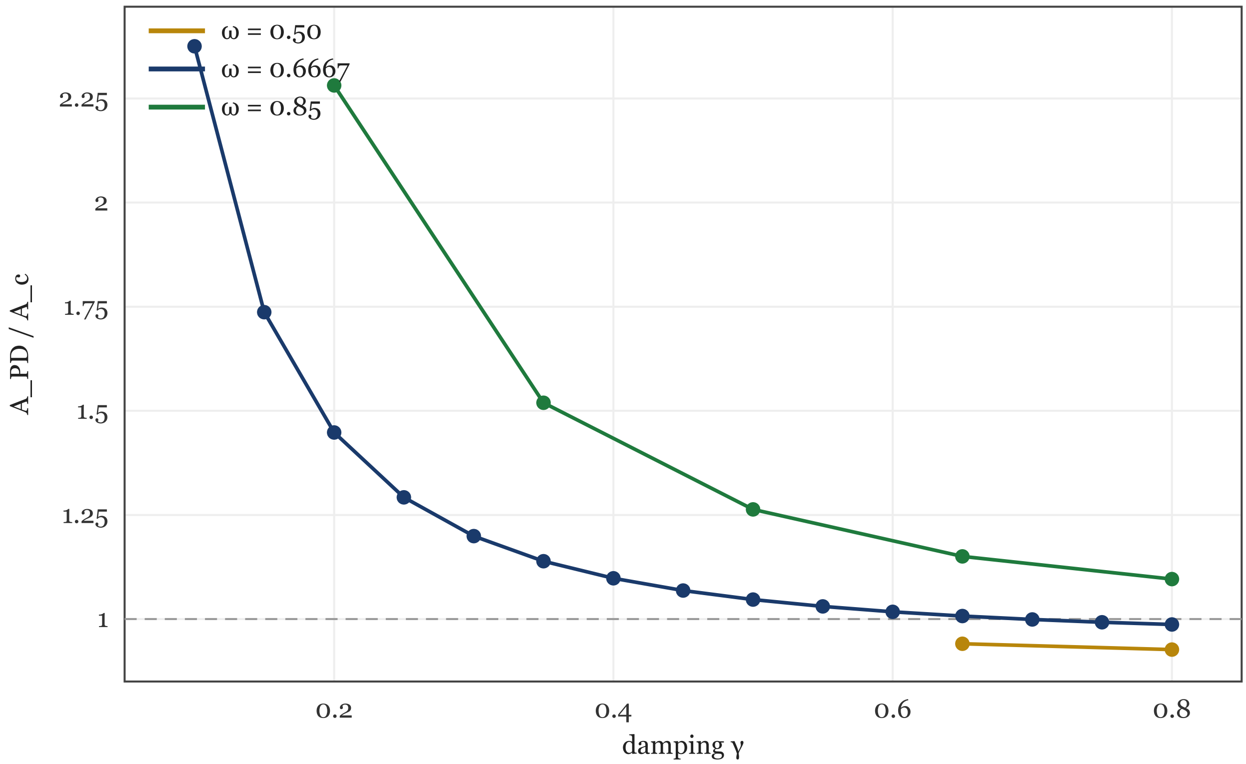

Figure 4. A_{PD}/A_c versus damping at three drive frequencies (Floquet-refined points only). The monotone closure persists at every ω, and the ratio-1 crossing shifts with frequency — the reversal is a property of the first-order threshold itself, not of ω = 2/3.

Second, the same question is posed to a different system: the symmetric **Duffing double well** $\ddot{x} = -\delta\dot{x} + x - x^3 + \Gamma\cos(t)$. The closed-form Melnikov threshold Γ_c(δ) (derived and quadrature-verified in the engine, reducing to the Guckenheimer–Holmes expression at α = −1, β = 1) is compared with the marched/bisected loss of period-1 stability of the confined single-well attractor. There is no Newton–Floquet refinement for the Duffing flow yet, so Γ_PD is quoted as a bisection bracket midpoint — honestly wider than the pendulum's interpolated onsets, though the brackets themselves are ~10^{-8} wide.

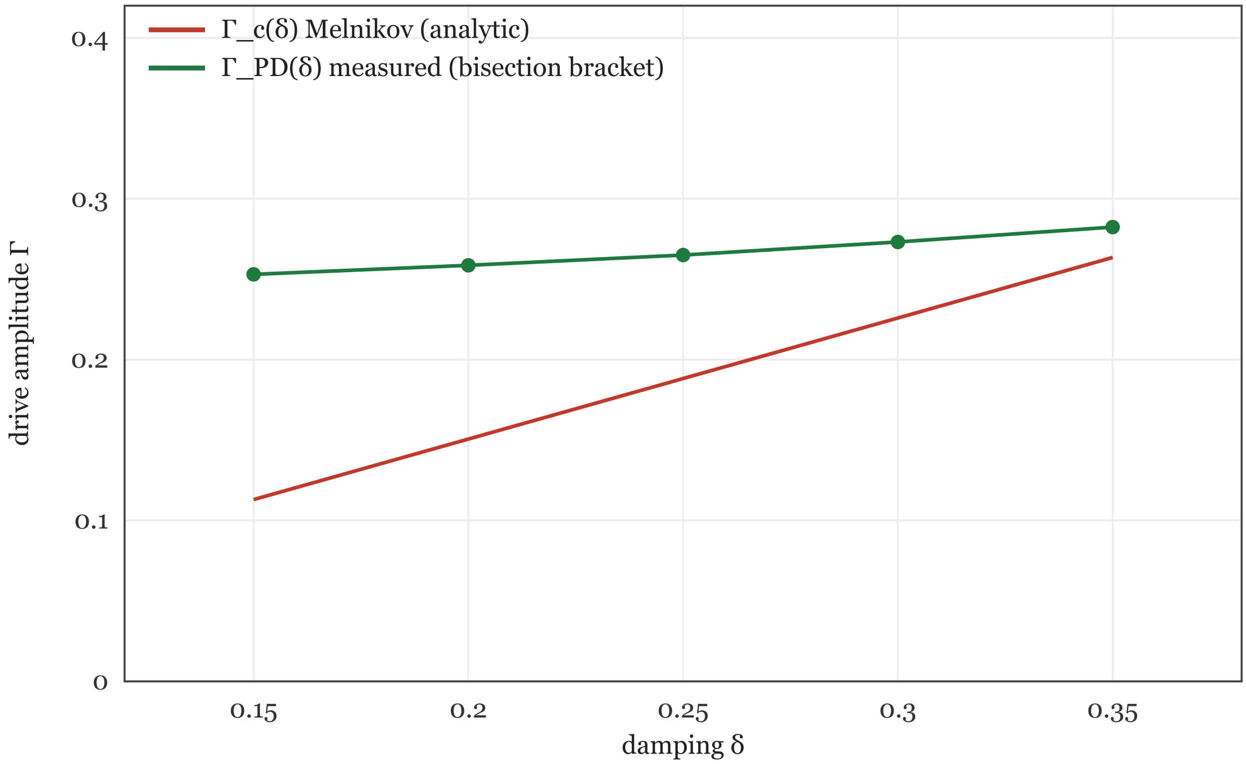

Figure 5. The Duffing double-well gap map: analytic $\Gamma_c(\delta)$ (red) versus the measured period-doubling onset $\Gamma_{PD}(\delta)$ (green). The ratio falls from 2.24 at δ = 0.15 to 1.071 at δ = 0.35 — the same monotone closure as the pendulum, in a system with a polynomial (non-pendulum) nonlinearity.

δ	Γ_c (Melnikov)	Γ_PD (measured)	δΓ_PD	Γ_PD/ Γ_c	K at 0.97·Γ	K at 1.08·Γ
0.15	0.11295	0.25301	1.7e-8	2.2400	-0.01	1.00
0.20	0.15060	0.25861	2.3e-8	1.7172	0.03	1.00
0.25	0.18825	0.26498	2.9e-8	1.4076	0.02	1.00
0.30	0.22591	0.27307	3.4e-8	1.2088	-0.02	1.00
0.35	0.26356	0.28232	4.0e-8	1.0712	-0.02	1.00

The 0–1 test corroborates every Duffing onset (K ≈ 0 below, K ≈ 1 above — the cascade accumulates quickly in the double well at these parameters). Together the two extensions upgrade the paper's claim from "the thresholds cross at one classic parameter point" to "first-order Melnikov ordering degrades with damping across drive frequencies and across systems".

6. Limitations and reproducibility

The main grid fixes ω = 2/3 (the frequency scan of §5 samples ω = 0.5 and 0.85 on a reduced γ grid) and follows a single attractor branch per γ (warm-started in A); coexisting attractors reached from other initial conditions may double elsewhere. At the lowest dampings the phase space is multistable enough that a finer warm-started march can hop basins before the doubling — at $\gamma = 0.15$ a basin-capture transition of the followed state was observed near $A \approx 0.49$ (the orbit itself remains strongly stable there, $\rho \approx +0.29$), below the verified doubling at 0.531. The quoted A_{PD} values are therefore specifically the $\rho = -1$ doublings of the primary oscillating branch, not necessarily the first event of any kind along a slow amplitude sweep. The Melnikov comparison concerns the first-order formula specifically — higher-order or numerical manifold computations would move A_c. The full study regenerates with `npm run paper:study` (~8 min) followed by `npm run flagship:certify`, `npm run flagship:external`, and `npm run paper:build`; the engine tests and independent symbolic/trajectory validations are in the same repository.

Appendix A. Certified onset localization

The flagship claim is not that Melnikov theory predicts the attractor cascade. It is a measured separation map: A_c is analytic first-order geometry, A_PD is a Floquet-refined attractor-branch instability, and the reported reversal is bounded by the exported caveat map and the independent Python A_PD probes. The ratio crossing is localized to $\gamma \in [0.69296963, 0.69297705]$, with point estimate 0.69297334. The Figure 1 hash is `07f877d6fdb816`.

γ	A_c	A_PD	δA_PD	A_PD/A_c	ρ below	ρ above	caveat
0.10	0.203755	0.483984	1.24e-7	2.375324	-0.94620	-1.05181	none
0.15	0.305632	0.530802	1.87e-7	1.736736	-0.94374	-1.05618	none
0.20	0.407510	0.590045	2.49e-7	1.447928	-0.94028	-1.06052	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.25	0.509387	0.658288	3.11e-7	1.292314	-0.93675	-1.06459	none
0.30	0.611265	0.732944	3.73e-7	1.199062	-0.93298	-1.06741	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.35	0.713142	0.812174	4.35e-7	1.138868	-0.93122	-1.07077	none
0.40	0.815019	0.894700	4.97e-7	1.097765	-0.92990	-1.07295	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.45	0.916897	0.979636	5.60e-7	1.068426	-0.92844	-1.07334	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.50	1.018774	1.066373	6.22e-7	1.046721	-0.92894	-1.07417	none
0.55	1.120652	1.154485	6.84e-7	1.030191	-0.92980	-1.07403	none
0.60	1.222529	1.243682	7.46e-7	1.017303	-0.93128	-1.07337	none
0.65	1.324407	1.333771	8.08e-7	1.007071	-0.93336	-1.07236	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD

γ	A_c	A_PD	δA_PD	A_PD/A_c	ρ below	ρ above	caveat
0.70	1.426284	1.424635	8.71e-7	0.998844	-0.93602	-1.07112	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.75	1.528161	1.516220	9.33e-7	0.992186	-0.94002	-1.07052	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.80	1.630039	1.608533	9.95e-7	0.986807	-0.93970	-1.06492	none

Appendix B. Independent Python A_{PD} reproduction

Python stdlib recomputation of A_c, crossing arithmetic, and selected A_PD values via RK4 strobe-map Newton/Floquet search. The analytic threshold agrees with maximum absolute error 0.00e+0. All selected period-doubling checks passed: **yes**.

γ	reported A_PD	Python A_PD	|Δ|	ρ low	ρ high	status
0.50	1.06637286	1.06637292	5.92e-8	-0.9999996	-1.0000004	pass
0.65	1.33377104	1.33377109	4.89e-8	-0.9999995	-1.0000003	pass
0.70	1.42463496	1.42463501	4.84e-8	-0.9999996	-1.0000004	pass

Independent-check caveat. A_PD checks use finite-difference monodromy and a coarser dt than the TypeScript flagship run, so they certify independent reproducibility at reviewer-kit tolerance, not bitwise equality.

Appendix C. Artifact integrity and caveat ledger

Artifact	Reproduce	Cross-reference
`reports/paper-study.json`	`npm run paper:study`	`Figures 1-3, main table`
`reports/flagship-certification.json`	`npm run flagship:certify`	`Figure 1 hash 07f877d6fdb816; Appendix A`
`reports/flagship-external-check.json`	`npm run flagship:external`	`Appendix B`
`paper/paper.pdf`	`npm run paper:build`	`This manuscript`

Caveat map. gamma=0.20: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD gamma=0.30: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD gamma=0.40: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD gamma=0.45: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD gamma=0.65: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD gamma=0.70: post-onset 0-1 sample is not cleanly chaotic at

1.08*A_PD gamma=0.75: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD Error bars combine attractor-bracket width with the available dt-sensitivity probe; they are a localization contract, not a full Bayesian posterior. Basin caveats are inferred from the exported 0-1 strobe probes; they flag multistability/transient-chaos risk but do not replace a full basin scan.

References

G. L. Baker and J. P. Gollub, *Chaotic Dynamics: An Introduction*, 2nd ed., Cambridge University Press (1996).

J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, Springer, §4.5 (1983).

V. K. Melnikov, “On the stability of the center for time-periodic perturbations,” *Trans. Moscow Math. Soc.* **12**, 1–57 (1963).

G. A. Gottwald and I. Melbourne, “On the implementation of the 0–1 test for chaos,” *SIAM J. Appl. Dyn. Syst.* **8**, 129–145 (2009).

M. J. Feigenbaum, “Quantitative universality for a class of nonlinear transformations,” *J. Stat. Phys.* **19**, 25–52 (1978).